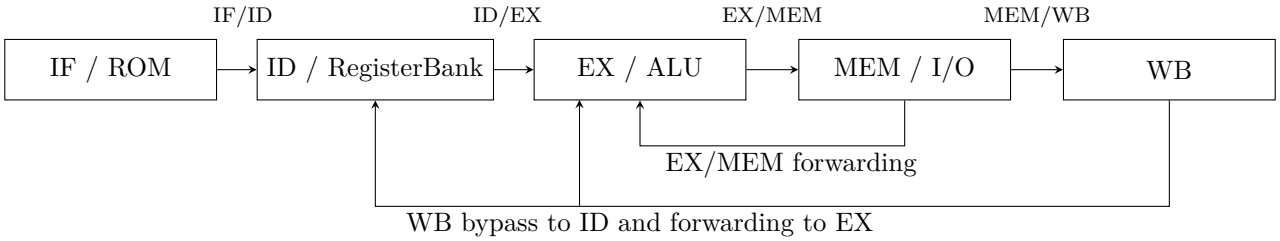
Architecture and control

Five-Stage Pipelined Microcontroller for a Washing Machine
Logisim-Evolution 5.0.0 • Assignment 2



1 Datapath and hierarchy

Main is the compact operator view. Its global Clock and sensor pins connect by explicit wires to WashController. The core contains Processor, the preloaded 256 x 16 PROGRAM_ROM, and MemoryAndIO. MemoryAndIO contains 128 x 8 DATA_RAM, the output latches, Timer and DoorSafety. There are no tunnels anywhere.

Only gates, splitters, pins, a clock, ROM, RAM and individual Registers are used. ByteAdder is an eight-bit ripple carry network of XOR/AND/OR gates. MUX2/3/4/7/8/16, EQ2/3/4/7/8/16, ALU, decoder and register selection are gate-built subcircuits; no built-in arithmetic/selection component is instantiated.

2 Actual pipeline-register fields

Bank	Fields	Bits
IF/ID	valid, PC[7:0], IR[15:0]	25
ID/EX	valid, PC[7:0], Rd[2:0], Rs[2:0], IMM[7:0], OP[3:0], A[7:0], B[7:0], RW/LD/ST/UD/US/BZ/BN/JP/HT	52
EX/MEM	valid, PC[7:0], Rd[2:0], result/address[7:0], store data[7:0], RW/LD/ST	31
MEM/WB	valid, PC[7:0], Rd[2:0], selected writeback data[7:0], RW	21

Banks are separate built-in Registers with a common rising-edge clock and reset, rather than packed wide registers. PC, HALTED, the eight general registers and peripheral state are additional state. Debug PCs travel through all four banks. VALIDS[3:0] = {WB, MEM, EX, ID}; forwarding selectors are 0=RF, 1=EX/MEM, 2=MEM/WB.

3 Forwarding, stalls, redirects and retirement

ID reads are gate-selected from R0-R7. A same-cycle WB match overrides each ID read, making WB writes visible to ID before the next edge. In EX, both operand A (Rd) and operand B (Rs) have an independent forwarding unit. Store data is forwarded A; its address is forwarded B.

Youngest priority: match = valid AND RegWrite AND destination=source, qualified by source-use. EX/MEM wins when it is not a load. MEM/WB is selected only if there is no EX/MEM match; an unready younger load must not expose a stale older value. Load-use interlocking keeps the consumer out of EX until MEM/WB can supply the load.

Stall: ID/EX.valid AND ID/EX.LD AND IF/ID.valid AND ((ID.UD AND ID.Rd=EX.Rd) OR (ID.US AND ID.Rs=EX.Rd)). Freeze PC and IF/ID for one cycle; write ID/EX.valid=0. EX/MEM and MEM/WB continue advancing, so a MEM store cannot repeat during a younger stall.

Priority	Action at the edge
1 Reset	PC=0; valid bits=0; HALTED=0; all latches/timer/trip cleared. This implementation also clears general registers. RAM is not cleared.
2 EX redirect / HALT	Taken branch uses forwarded A. Redirect selects absolute IMM; IF/ID and ID/EX invalidated. Valid HALT latches HALTED and kills younger entries.
3 Load-use stall	Hold PC, IF/ID; inject one invalid ID/EX bubble.
4 Normal	Fetch PC, PC+1, advance all pipeline banks. Older entries always advance after reset is released.

A redirect has priority in PC and IF/ID enable logic; branch/HALT flush has priority over inserting a valid ID/EX entry. With this single-EX-stage design, an EX load and EX branch cannot both be valid simultaneously. The priority logic remains explicit.

Writes: WBWE = MEMWB.valid AND RW AND NOT reset. MEMSTORE = EXMEM.valid AND ST AND NOT reset. RAM select is NOT address[7]. Peripheral writes use exact-address gate comparators. Unimplemented/reserved reads are zero, and their writes are ignored. Read-only writes are ignored.

HALT: a valid EX HALT stops fetch immediately, while older MEM/WB entries retire. Invalid/wrong-path HALT cannot stop the Processor. Timers continue to tick after halt. The wash program explicitly disables its timer before completion.

4 Timer and mandatory safety interlock

Timer has COUNT[7:0], LIMIT[7:0], ENABLE and sticky MATCH registers. A custom ByteAdder computes COUNT+1; a gate comparator checks the incremented count against the old nonzero LIMIT. The timer shares the global clock, not Processor pipeline enables. Its next state uses priority reset > software clear > enabled increment > hold. Clearing resets COUNT and MATCH together. LIMIT writes do not retroactively compare the old count.

Physical output: ACTUAL_MOTOR = CPU_MOTOR_EN AND DOOR_CLOSED AND NOT TRIP. This is combinational, so opening the door disables the motor without waiting for a clock or software polling. TRIP sets at a rising edge if the motor request is active and the door is open; it stays set after the door closes.

START_PREV samples START every clock. A qualified rising edge (door closed and motor request already zero) sets PENDING. A valid MEM write of bit 0=1 to FA accepts START only if START is still high, the door is closed, motor request is zero, and PENDING or a newly qualified edge exists. Acceptance captures MODE, clears TRIP and consumes PENDING. A trip clears pending edges. TRIP set has priority over acceptance clear; reset has highest priority.

FC bit 0 reports whether the last FA write succeeded; bit 1 reports TRIP. Failed acknowledgement does not clear TRIP. Software aborts a detected motor-phase trip by clearing the actuator request, disabling the timer, displaying EE and returning to WAIT_START. A fresh safe START transition is required to begin again. Normal completion halts and requires reset for a new cycle.

5 Washing program

Parameter	**MODE 00**	**MODE 01; also 10/11 fallback**
Fill target	100 (64 hex)	150 (96 hex)
Wash timer limit	80 (50 hex)	120 (78 hex)
Spin timer limit	40 (28 hex)	60 (3C hex)
Drain threshold	<= 10	<= 10

MODE is captured by successful FA acceptance, then read at FB; subsequent MODE-pin changes do not affect the cycle. Registers R2/R3/R4 retain target/wash/spin parameters. R7=3 masks START/DOOR, R5=2 masks TRIP, and R0/R1/R6 are initialized temporaries. The program never relies on uninitialized RAM or general registers.

Sequence: wait/accept -> lock and fill -> close valve -> clear/rearm wash timer -> alternate motor direction -> stop motor and pump -> drain <=10 -> pump off -> clear/rearm spin timer -> spin -> motor off -> timer off -> display AA -> buzzer with door unlocked -> HALT. TEMP and RPM are readable sensor-only interfaces.

This is the educational digital interlock required by the assignment, not a certified appliance safety system.